

15 B Work done on gas = $25 \times 10^3 \times (80 - 40) \times 10^{-3} = 1000 \text{ J}$
 $\Delta U = Q + W = -2500 + 1000 = -1500 \text{ J}$
 Avoid using $\Delta U = \frac{3}{2} \Delta(PV)$ as the gas may not be monatomic.

16 A I